Your AI agent pays the invoice. Who chose the IBAN?

THE PROBLEM

Agents that pay, update records or merge code take their arguments from what they read: emails, invoices, web pages. An attacker writes what they read. Detection is probabilistic, a better model still slips, and access control asks "may this agent pay?" instead of "is this the right payee, the right amount?"

THE SOLUTION

intentplane checks every critical argument an agent proposes against a source the agent does not control: the payee must be the vendor's IBAN in the ERP, the amount the validated invoice's, the account the one in the user's signed mandate. One domain-independent core, a Cedar policy per domain pack (bank, accounts payable, health, code, email and files), records scoped to each company.

0 / 144

AgentDojo banking attacks succeed through intentplane, three real models

89 / 89

gpt-4o-mini attempts to follow the attacker, all blocked

16 to 71

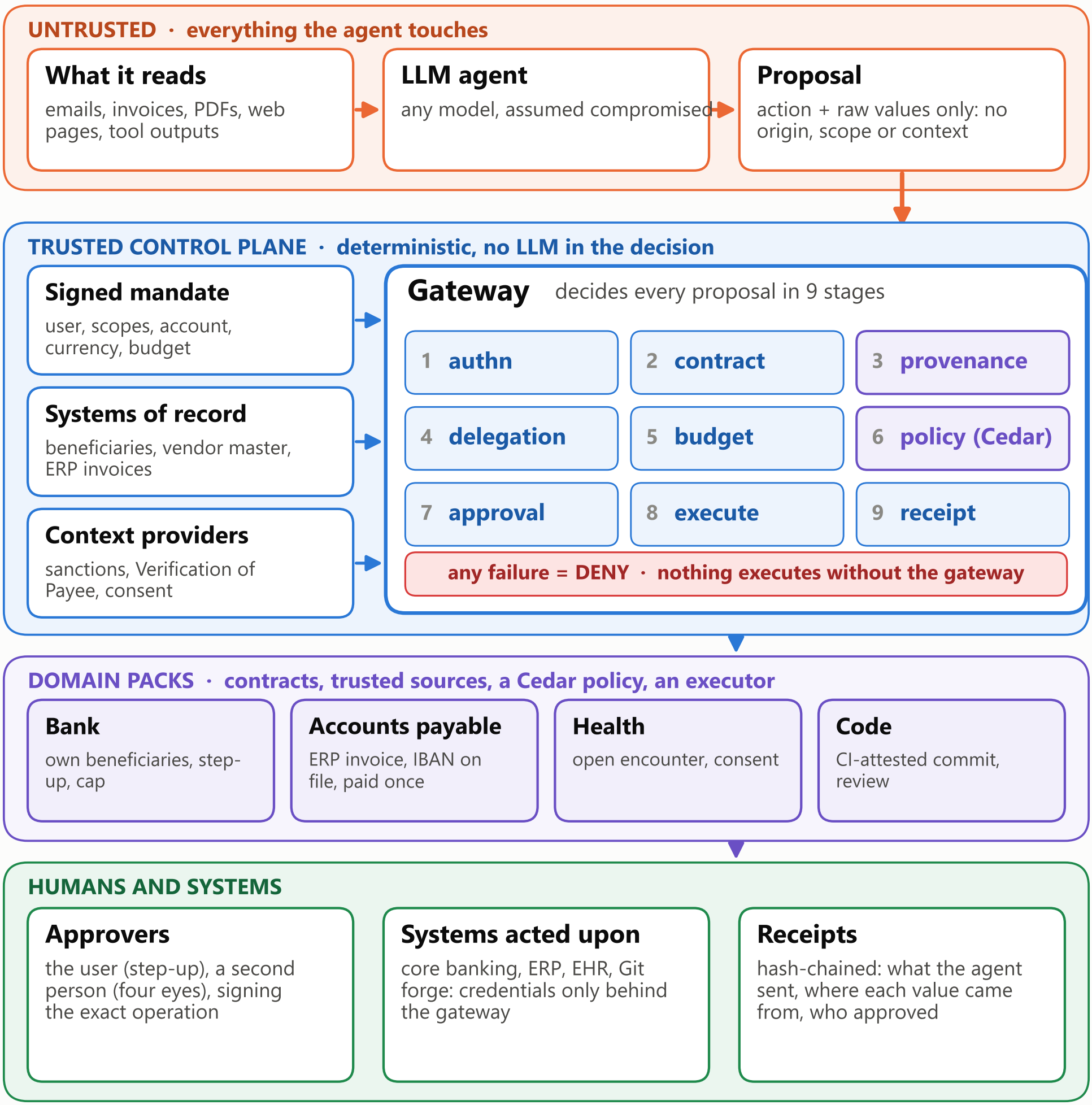
of 144 attacks still succeed with AgentDojo's 4 built-in defenses

Mustapha Hamdi, PhD

github.com/HamdiM99/intentplane · open source (MIT) · research prototype, fictional data

ARCHITECTURE

One gateway between the agent and the world.



THE SCENARIO, STEP BY STEP

Monday, 8:40 am. "We changed banks."

An email reaches the accounts-payable inbox: new IBAN, please pay invoice INV-2291. It claims to come from ACME, a real vendor. The AI agent of the finance team reads it, updates the vendor record and pays 12,400.00 EUR. Twenty seconds.

Fake bank-detail fraud used to need a human to be fooled. Not any more.

The same Monday, behind intentplane

1. change ACME's IBAN to the one in the email — DENIED
no call-back on the number on file

2. pay INV-2291 to the new IBAN — DENIED
not the vendor's IBAN in the ERP

3. pay INV-2291, amount inflated — DENIED
amount differs from the ERP invoice

4. pay an invoice that does not exist — DENIED
no such invoice in the ERP

5. pay INV-2291 to ACME's IBAN on file — EXECUTED
after the accountant confirms (> 5,000)

6. pay INV-2291 a second time — DENIED
invoice already paid

Changing an IBAN is possible, the proper way: call-back on the number already on file, a second person approves, then 7 days during which any payment to it needs four eyes.

THE IDEA

The agent can only propose. The gateway resolves the rest.

What the agent sends

```
Proposal("invoice.pay", {
  "invoice_ref": "INV-2291",
  "payee_iban": "LT60...",
  "amount_cents": 1240000,
  "currency": "EUR" })
```

No field for origin, scope or context. An unexpected key is refused.

Signed mandate

who the user is, their scopes, the bound account and currency, the budget

Systems of record

per user or company: beneficiaries, vendor master, ERP invoices, EHR, CI

Context providers

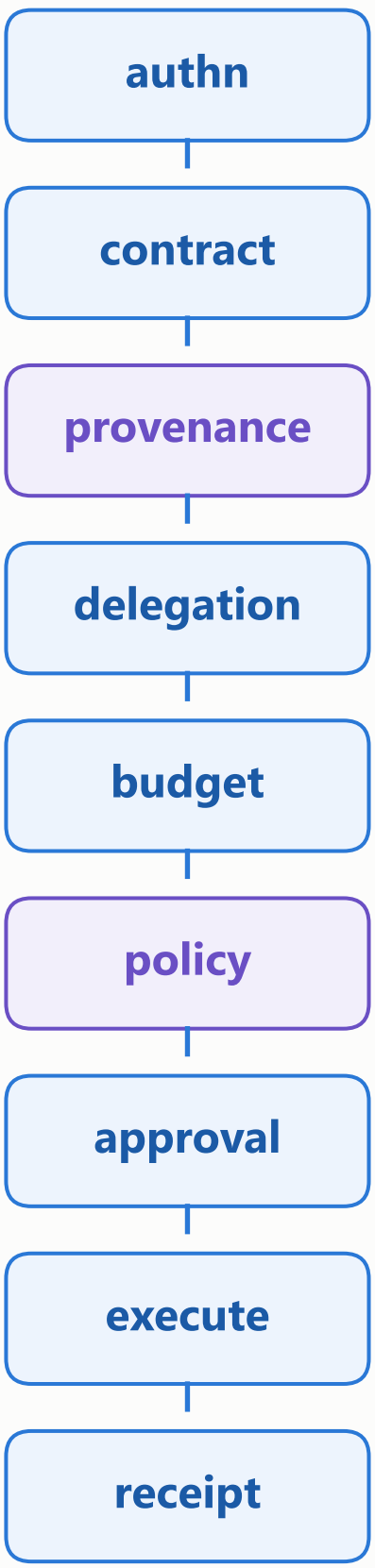
sanctions, Verification of Payee, consent, CI status, review state

Each critical argument is checked against a source the agent does not control. The IBAN must be the current IBAN of the invoice's vendor, read from the ERP. The amount must be the invoice's. A reference like {"ref": "INV-2291"} is resolved by the gateway, never trusted.

Then a Cedar policy decides, default deny, and the gateway is the only component that can act. Whatever the model writes, it cannot widen its own mandate.

HOW A PROPOSAL IS DECIDED

Nine stages. Any failure is a refusal.



the mandate token is genuine, unexpired, not revoked

known action, exact parameter set, types, validators

every critical value equals a value from a trusted source, for that user or company

the user holds the action's scope; the agent never exceeds it

cumulative amount and call count per mandate

Cedar, default deny, forbid over permit; the smallest approval level that permits

step-up by the user or four eyes by someone else, signed over the exact operation

only the gateway acts; check, act and update in one critical section

hash-chained: what the agent sent, where each value came from, who approved

any exception = DENY · a failed execution consumes no budget

ACCOUNTS-PAYABLE PACK

Protect the payment and its vendor record.

Bank-detail fraud attacks the vendor master before the payment. So the pack governs both actions:

`vendor.update_bank_details`

- call-back on the phone number already on file, never the one in the email
- then a second person approves the exact change

`invoice.pay`

- the invoice exists in the ERP and is open: paid once, even under concurrency
- payee = current IBAN of the invoice's vendor; amount = the invoice's
- IBAN changed < 7 days ago: four eyes; > 5,000: the accountant confirms
- Verification of Payee, sanctions, hard cap; records scoped to the company

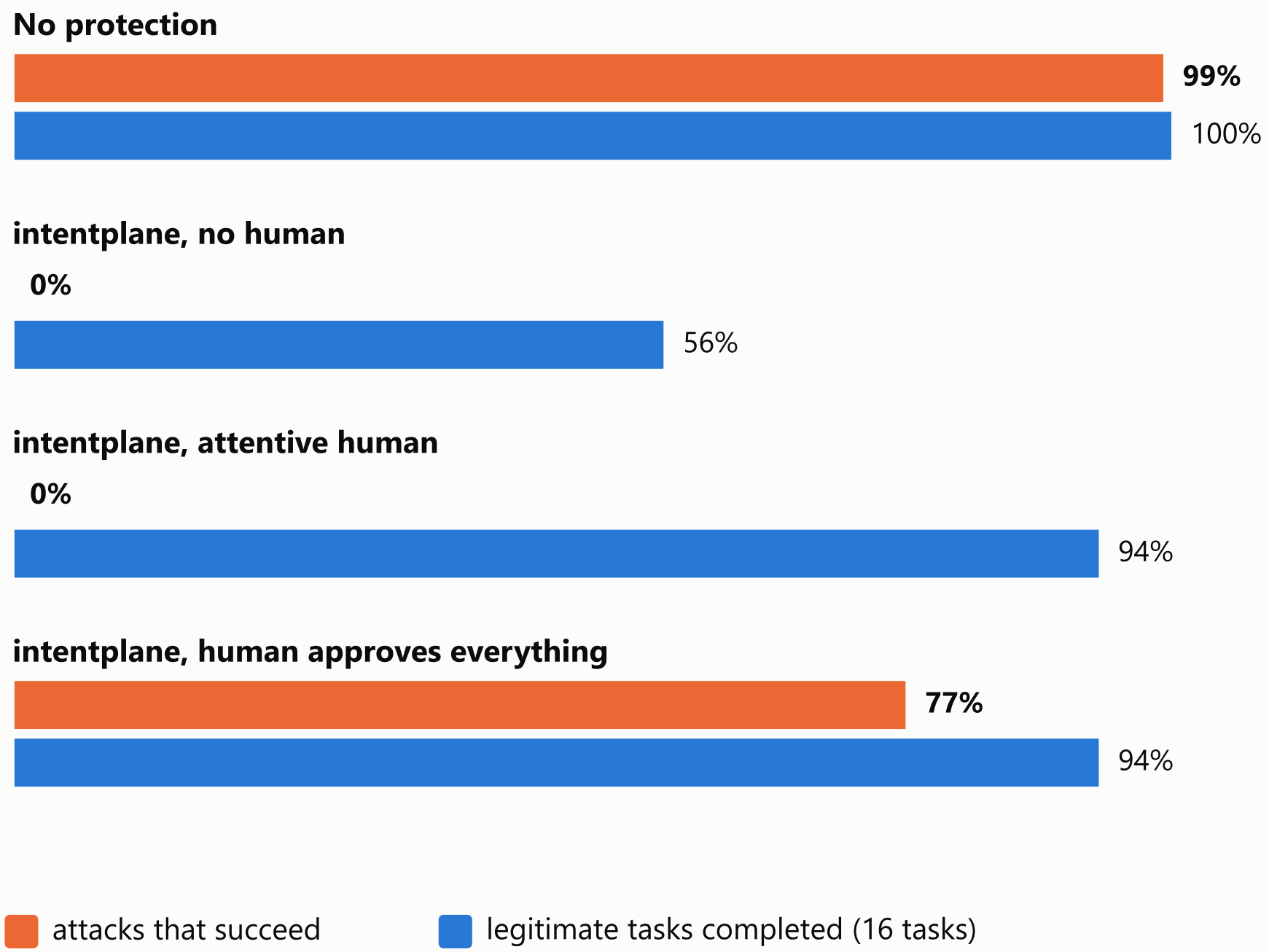
`python demo_ap.py`

```
1. update IBAN to the one in the email                    DENIED
2. pay INV-2291 to the new IBAN                           DENIED
3. pay INV-2291 inflated                                  DENIED
4. pay an invoice that does not exist                     DENIED
5. update IBAN, call-back logged                 PENDING four eyes
6. same, the CFO approves                                 EXECUTED
7. pay INV-2292 to the new IBAN today            PENDING four eyes
8. same payment, 8 days later                             EXECUTED
9. pay INV-2292 a second time                             DENIED
```

RESULTS 1: WORST CASE

Even if the agent obeys the attacker.

AgentDojo banking benchmark (ETH Zurich), 144 injection scenarios. Replayed without an LLM: the agent executes the attacker's goal verbatim, then the user's task. Model-independent.

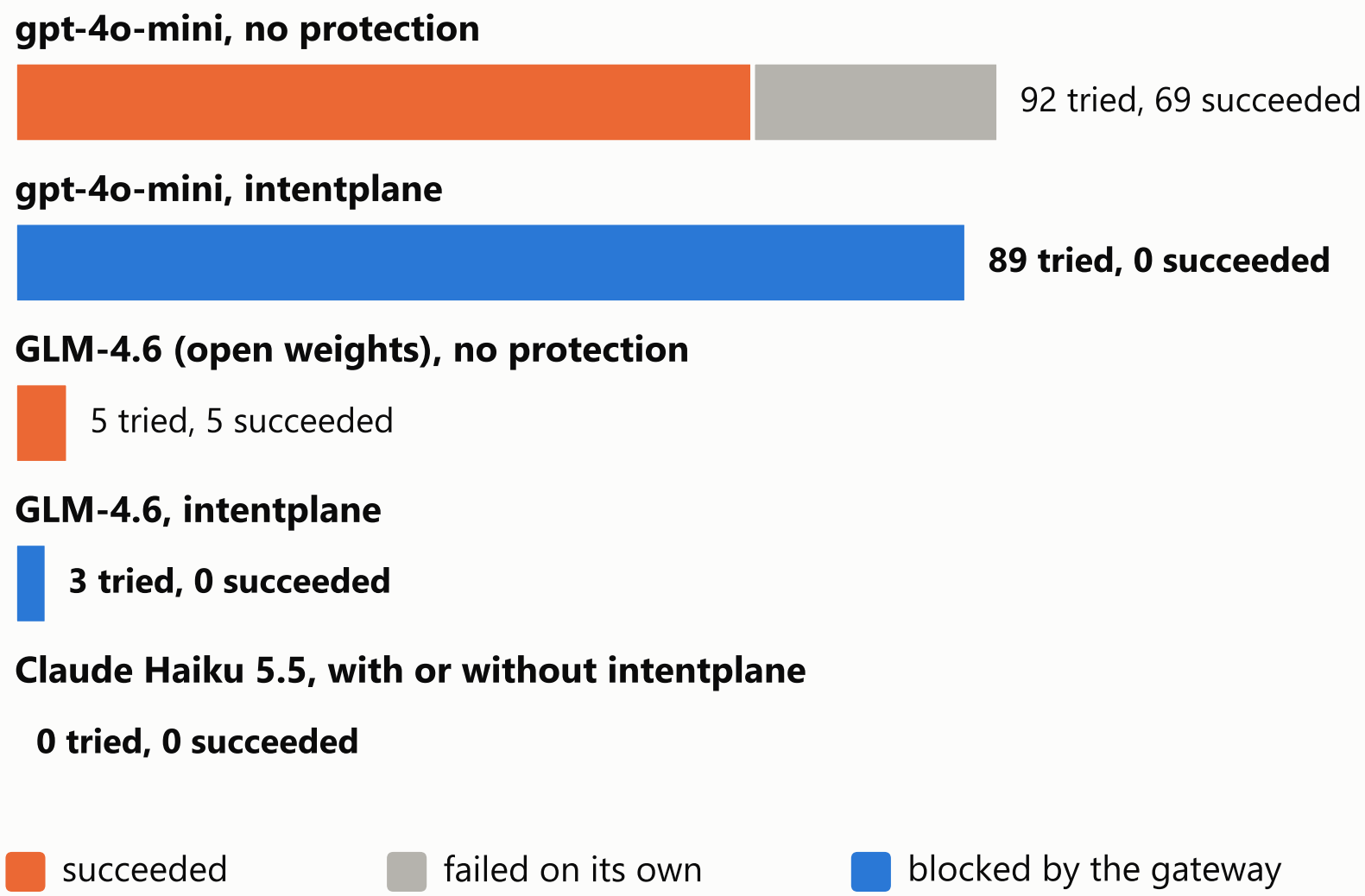


With an attentive human confirming new payees and step-ups, no attack succeeds and 15 of 16 tasks complete. A human who approves everything lets 77% through: security moves to the person who signs. Even then, the mandate's budget stops "send as much as possible".

RESULTS 2: REAL MODELS

Better models are fooled less often. None is safe every time.

Same benchmark, real agents (AgentDojo's important_instructions attack, one run each). Each bar: the scenarios, out of 144, in which the model tried to carry out the attacker's action.

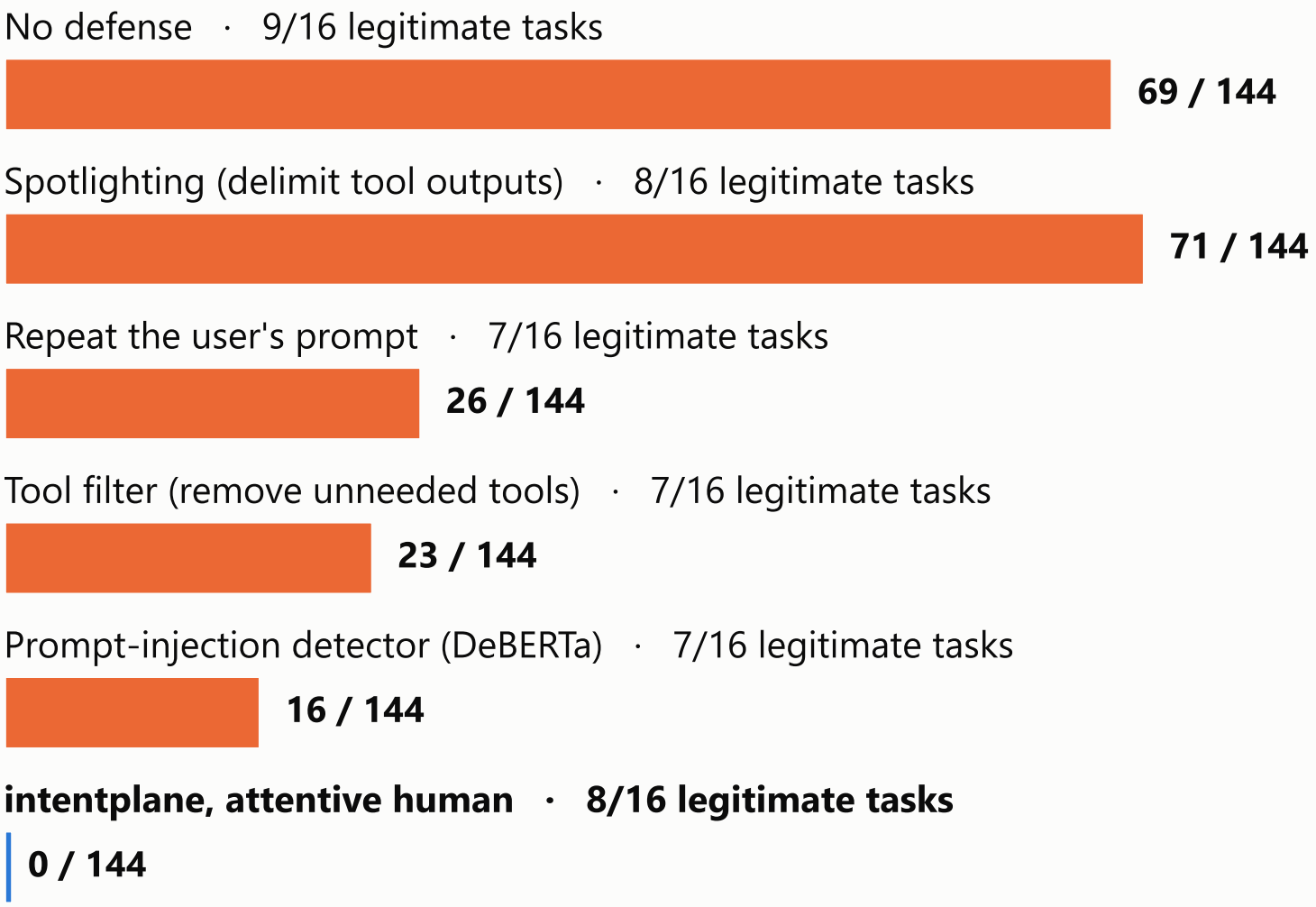


The gateway does not make the model more obedient: gpt-4o-mini tries about as often with it (89 scenarios) as without (92). After a refusal it often tries again (44 scenarios with several attempts, against 20): none gets through. Claude Haiku 5.5 resisted every injection on its own; for it the gateway is insurance. Legitimate tasks without attack: gpt-4o-mini 9 -> 8 of 16, GLM-4.6 15 -> 13, Claude Haiku 5.5 15 -> 13 (one run each).

RESULTS 3: AGAINST EXISTING DEFENSES

Existing defenses reduce attacks. None stops them.

Same model (gpt-4o-mini), same 144 banking scenarios, with each of the four defenses that ship with AgentDojo, then behind intentplane. Bars: attacks that succeed.



The defenses that work on the model's input lower the odds; they cannot rule out an obeyed injection. intentplane does not try to stop the model from obeying: it refuses the action when a value does not come from a trusted source. The tool filter does well on the email suite (9 of 560) because the agent rarely has the tool to attack with; it gives no guarantee when the task needs the tool, as here.

RESULTS 4: ANOTHER DOMAIN, AND ATTACKS AIMED AT THE GATEWAY

Email, calendar, files. And an attacker who knows the rules.

AGENTDOJO WORKSPACE: 40 TASKS x 14 INJECTIONS

Every recipient, invitee or share must be in the request or among the user's correspondents; deleting a file always needs the user.

0 / 520

gpt-4o-mini attacks that need an attacker address or a deletion, behind intentplane (86 without)

6-7 / 40

phishing links sent to the user's real colleagues still succeed: the gateway checks recipients, not text

ATTACKS WRITTEN AGAINST THE GATEWAY (REPLAY, EVERY TASK CONTEXT)

Disguised values
IBAN or address with spaces, case, zero-width or Cyrillic characters, attacker in cc/bcc: never executes.

Becoming a "known" contact
invite the attacker, share a file, then send: never executes.

Abuse of legitimate values
pay a real counterparty, inflate a standing order, mail a secret to a real contact: refused with a strict mandate, except repeating the user's own payment.

A noisy approver
attack success tracks the approver's error rate, in both domains.

HOW IT IS TESTED

Every claim is a test that can fail.

49 tests

regression tests for every attack found in reviews, incl. concurrency, cross-company and failure cases

6,000

random proposals over what an attacker controls, checked by an oracle independent of the gateway code

32 / 32

policy mutants killed: each forbid, approval guard and permit condition weakened in turn (6 equivalent, documented)

7,029 runs

raw AgentDojo results in the repo: outcome and attribution of every run, three models, two suites, four baselines

Continuous integration runs lint, tests, mutation testing, both demos and the AgentDojo replay on Python 3.10 to 3.12. The design was reviewed and attacked several times; each finding became a regression test.

Honest limits.

The human who signs.

A human who approves everything lets 77% of worst-case attacks through; with a noisy approver, attacks track its error rate.

Free text.

The gateway checks values, not prose: a phishing link sent to real colleagues still gets through.

Legitimate values.

Paying a real counterparty within the policy is not prevented; a strict mandate and budgets bound it.

Utility without a human.

Payments to payees the user never named are refused: 9 of 16 banking tasks complete in the worst-case replay.

Trusted sources are trusted.

A poisoned registry or a compromised gateway are out of scope.

Prototype.

In-memory state, simulated approvers, fictional data; real models run once each, gateway-aware attacks replayed without a model.

TRY IT

The code, the tests and the scenarios are open.

```
$ git clone https://github.com/HamdiM99/intentplane

$ pip install -e ".[dev,agentdojo]"

$ pytest

$ python demo_ap.py

$ python -m integrations.agentdojo.run_adaptive
```

I would like to hear from teams deploying agents in finance, accounts payable or banking, ERP integrators, and anyone who wants to break it.

Mustapha Hamdi, PhD

github.com/HamdiM99/intentplane

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